Experiment No:19
Experiment to understand Logistic Regression for a given data set

AIM:
To understand Logistic Regression for a given data set.

DESCRIPTION:
Understand the Logistic Regression algorithm for the dataset given.

CODE:

```
import numpy as np
import pandas as pd

df=pd.read_csv('/content/Breast_Cancer_Logistic_Regression.csv')

df.head()

features=df.iloc[:,:-1].values
label=df.iloc[:,-1].values

features
label

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

for i in range(1,401):

x_train,x_test,y_train,y_test=train_test_split(features,label,te
st_size=0.2,random_state=i)

    model=LogisticRegression(max_iter=10000)
```

```
    model.fit(x_train,y_train)

    train_score=model.score(x_train,y_train)
    test_score=model.score(x_test,y_test)

    if test_score>train_score:
        print("Test {} Train {} Random State
{}".format(test_score,train_score,i))

x_train,x_test,y_train,y_test=train_test_split(features,label,te
st_size=0.2,random_state=42)

finalModel=LogisticRegression(max_iter=10000)
finalModel.fit(x_train,y_train)

from sklearn.metrics import classification_report

print(classification_report(y_test,finalModel.predict(x_test)))
```

OUTPUT:

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Test 0.9649122807017544 Train 0.9538461538461539 Random State 5
Test 0.9649122807017544 Train 0.9560439560439561 Random State 8
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	precision	recall	f1-score	support
0	0.97	0.91	0.94	43
1	0.95	0.99	0.97	71
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

RESULT:

Thus, Logistic Regression was successfully implemented on the Breast Cancer dataset to classify the data into two categories. The dataset was divided into training and testing sets, and the model was trained using `LogisticRegression`. Its performance was evaluated using a classification report, achieving 96% accuracy on the test dataset.